Enterprise RAG Platform

Repository-based research report on grounded enterprise retrieval and answer generation.

Sai Veda

Independent Project Research

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1. Executive Abstract

This brief documents a retrieval-augmented generation core built for enterprise knowledge use cases where accuracy, traceability, and tenant boundaries matter more than surface-level demo output. The final stack turned retrieval into a measurable engineering system. Hybrid retrieval with reranking improved top-rank quality while preserving grounded, citation-backed answers and deterministic offline reproducibility. The implementation evidence is drawn from committed repository artefacts, benchmark outputs, automated tests, and generated screenshots.

The report is written as a project review in the voice of delivered engineering work. Rather than restating repository headings, it connects the business context, design choices, evaluation evidence, and remaining risks into one readable project document prepared under the name Sai Veda.

Keywords: Hybrid retrieval (BM25 + dense) + reranking + eval harness, implementation evidence, benchmark results, project validation

2. Operational Context

The operating problem was straightforward: teams keep policies, tickets, and wiki content in separate silos, but a useful assistant still has to retrieve the correct passage, cite it, and avoid leaking data across tenants. Production-grade retrieval-augmented generation - offline, deterministic, and evaluated.

2.1 Enterprise Search Problem Statement

The central project problem can be stated directly. The operating problem was straightforward: teams keep policies, tickets, and wiki content in separate silos, but a useful assistant still has to retrieve the correct passage, cite it, and avoid leaking data across tenants. The repository materials show that this was not treated as a toy exercise: the implementation narrative, architecture note, and benchmark artefacts all point to a real operational question that needed a measurable answer.

2.2 Review Questions And Working Assumptions

The document review was guided by a small set of concrete questions. Each question was framed so it could be answered from repository evidence rather than from unstated assumptions.

Research question: Does the implemented project address the stated technical problem in a complete and reviewable manner?

Hypothesis: The repository design, architecture notes, and implementation artefacts are expected to demonstrate end-to-end coverage of the core project objective.

Research question: Do the benchmark and test artefacts support the main performance or quality claim?

Hypothesis: The recorded evidence is expected to support the headline result reported for this project: Hybrid+Rerank: Recall@1 0.92, nDCG@10 0.96.

Research question: Can the results be reviewed and reproduced from committed repository evidence?

Hypothesis: The presence of 28 automated tests, benchmark outputs, and generated screenshots is expected to provide a transparent validation path.

3. Retrieval Design Basis

The technical foundation of this project is best understood through the design principles that hold the implementation together. Built around retrieval quality, citation traceability, and strict tenant isolation for enterprise knowledge systems.

3.1 Retrieval Design Basis

1. Hybrid fusion was kept score-scale agnostic so lexical and dense paths can coexist honestly. Built around retrieval quality, citation traceability, and strict tenant isolation for enterprise knowledge systems.

2. Citation grounding was treated as a product requirement, not a dashboard extra.

3. Isolation is applied inside the retriever to prevent ranking leakage and wasted top-k budget.

3.2 Repository Signals And Supporting Notes

Enterprise RAG Platform is a retrieval-augmented generation stack designed the way a production system is: an ingestion path that streams, an index layer with pluggable retrieval backends, a learned reranker, extractive synthesis with citations, an evaluation harness that gates quality, and hard multi-tenant isolation. Everything runs offline and deterministically so the whole pipeline is reproducible from a single seed. The encoder sits behind the Embedder protocol (fit / encode / dim), so a SentenceTransformerEmbedder could drop in without touching retrieval, fusion, reranking, or eval. That is the important architectural property: the offline model is an implementation detail, not a design constraint.

- Streaming corpus generation and chunking with metadata.
- BM25, dense, hybrid retrieval, and learned reranking.
- Extractive synthesis with citation offsets.

4. System Topology And Build Path

Built around retrieval quality, citation traceability, and strict tenant isolation for enterprise knowledge systems. The encoder sits behind the Embedder protocol (fit / encode / dim), so a SentenceTransformerEmbedder could drop in without touching retrieval, fusion, reranking, or eval. That is the important architectural property: the offline model is an implementation detail, not a design constraint. Trade-off: LSA has no external world knowledge, so it cannot resolve true synonyms the corpus never co-locates. In our benchmark this shows up as a strong lexical BM25 baseline that pure dense retrieval does not beat - an honest result. The value of the dense arm is robustness (it recovers documents whose exact query terms are missing) and feature diversity for the reranker.

4.1 Implementation Workflow

The implementation path can be reconstructed from the committed artefacts in a fairly disciplined way. Problem framing: Mapped enterprise search failure modes to measurable retrieval goals: recall, rank quality, groundedness, and tenant safety. Core build: Implemented chunking, BM25, dense retrieval, RRF fusion, and a logistic reranker behind clean interfaces. Validation: Benchmarked four retrieval paths on held-out questions and compared quality against latency and groundedness. Hardening: Added deterministic seeds, artifact generation, and multi-tenant tests so the same run can be reviewed repeatedly.

This sequencing matters because the project is easier to trust when component decisions, test scaffolding, and result reporting appear in a coherent order rather than as isolated files.

5. Evaluation Setup And Evidence Base

5.1 Repository Evidence Base

Source: committed repository artefacts including implementation code, automated tests, benchmark outputs, architecture notes, and generated visual evidence. Coverage: 28 automated tests, 1 benchmark table(s), and 4 screenshot asset(s) were available for document preparation. Schema / artefact types: README overview, ARCHITECTURE design note, benchmark markdown and CSV outputs, test suites, and figure assets generated from the project run path. Availability: all evidence cited in this document is sourced from the repository itself.

5.2 Evaluation Protocol

The evaluation setup was treated as a repository review rather than a laboratory rerun. Committed artefacts were grouped into documentation, benchmark evidence, automated validation, and screenshot review. In practical terms this meant examining 1 benchmark table(s), 4 screenshot asset(s), and 28 automated tests while reading the implementation narrative in sequence. The review lens stayed aligned with the project goal: Raise first-result relevance without hiding weak baselines. Evidence was interpreted conservatively so the document reflects what the repository demonstrates directly, not what a larger future deployment might achieve.

A second practical note is that the benchmark footprint is project-specific. The benchmark evidence reviewed for this report includes benchmarks/latency.csv, benchmarks/results.json, benchmarks/retrieval_quality.csv, benchmarks/retrieval_quality.md, benchmarks/scale.csv.

6. Findings And Interpretation

6.1 Benchmark Summary

The principal repository claim for this project is recorded as: Hybrid+Rerank: Recall@1 0.92, nDCG@10 0.96. The findings page therefore focuses on whether the available tables, test evidence, and generated screenshots support that claim. The final stack turned retrieval into a measurable engineering system. Hybrid retrieval with reranking improved top-rank quality while preserving grounded, citation-backed answers and deterministic offline reproducibility. The scorecards show Hybrid+Rerank leading both BM25 and dense-only retrieval on top-rank quality. Latency remains operationally plausible because the reranker only touches a bounded candidate set.

method	recall@1	recall@3	recall@5	recall@10	mrr
BM25	0.888	0.988	0.988	0.988	0.937
Dense	0.188	0.216	0.220	0.244	0.206
Hybrid	0.300	0.540	0.664	0.792	0.450
Hybrid+Rerank	0.920	0.984	0.988	0.988	0.952

6.2 Interpretation

The scorecards show Hybrid+Rerank leading both BM25 and dense-only retrieval on top-rank quality. Latency remains operationally plausible because the reranker only touches a bounded candidate set. Groundedness stays stable across retrieval modes because synthesis remains extractive and citation-aware.

The benchmark table is not treated as a marketing surface. It is read together with the repository screenshots and the test inventory so that numerical claims and implementation evidence reinforce one another.

7. Visual Validation Plates

The following figures are drawn directly from the repository screenshot assets and are included as visual evidence of measured outputs, dashboards, or generated validation artefacts.

7.1 Retrieval quality - 4 methods compared

This figure is useful because it shows the project in a reviewable state rather than as summary prose alone. The scorecards show Hybrid+Rerank leading both BM25 and dense-only retrieval on top-rank quality.

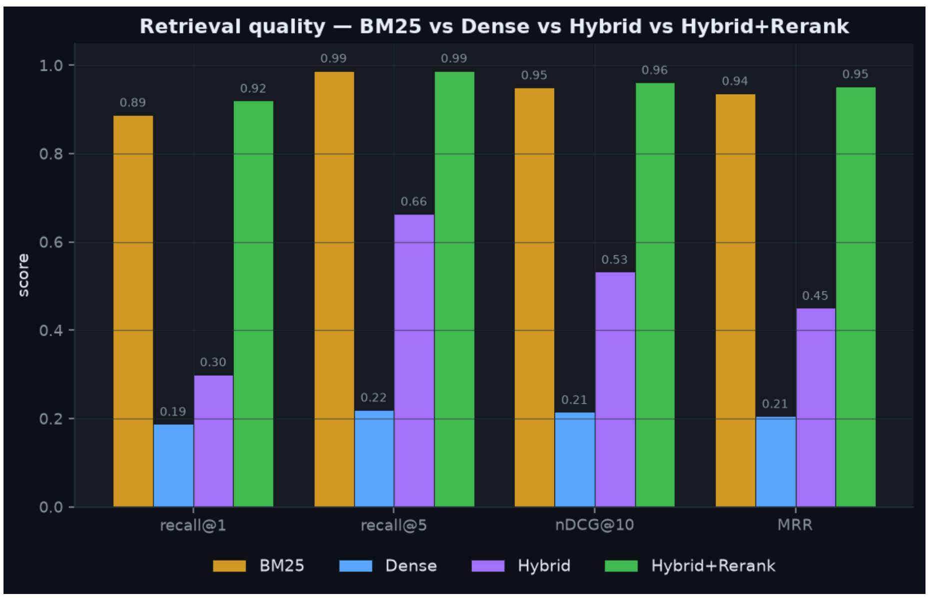

Figure 7.1. Retrieval quality - 4 methods compared

7.2 Evaluation scorecard

The visual evidence attached to Evaluation scorecard helps connect the quantitative story to the implemented workflow. Latency remains operationally plausible because the reranker only touches a bounded candidate set.

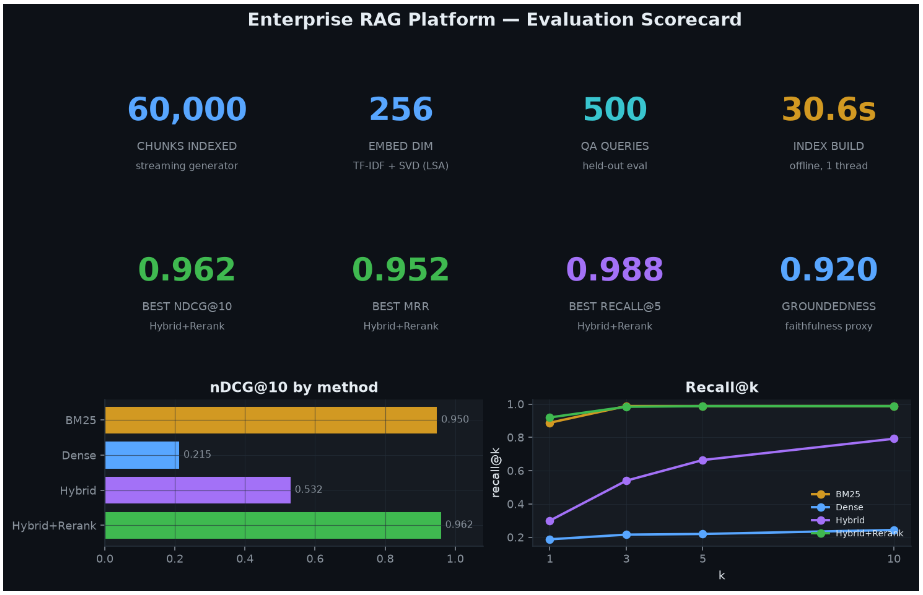

Figure 7.2. Evaluation scorecard

8. Validation Checklist And Residual Review

The second figure set focuses on validation continuity. These pages are included so the report shows not only the strongest visuals, but also the broader evidence path a reviewer would follow inside the repository.

8.1 Query latency vs retrieval depth

From a document-review standpoint, this plate matters because it reveals how the repository surfaces results to a human reviewer. Groundedness stays stable across retrieval modes because synthesis remains extractive and citation-aware.

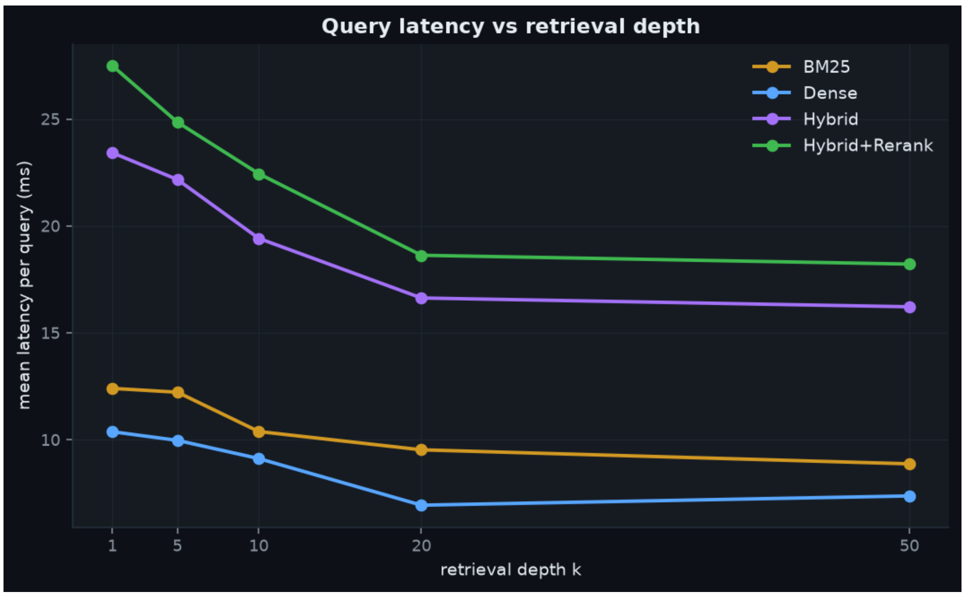

Figure 8.1. Query latency vs retrieval depth

8.2 Reranker lift & ranking quality

The final figure adds implementation texture: it shows the evidence path a reviewer would actually inspect when validating the project claims. Mitigated by applying tenant masks before scoring rather than filtering results after ranking.

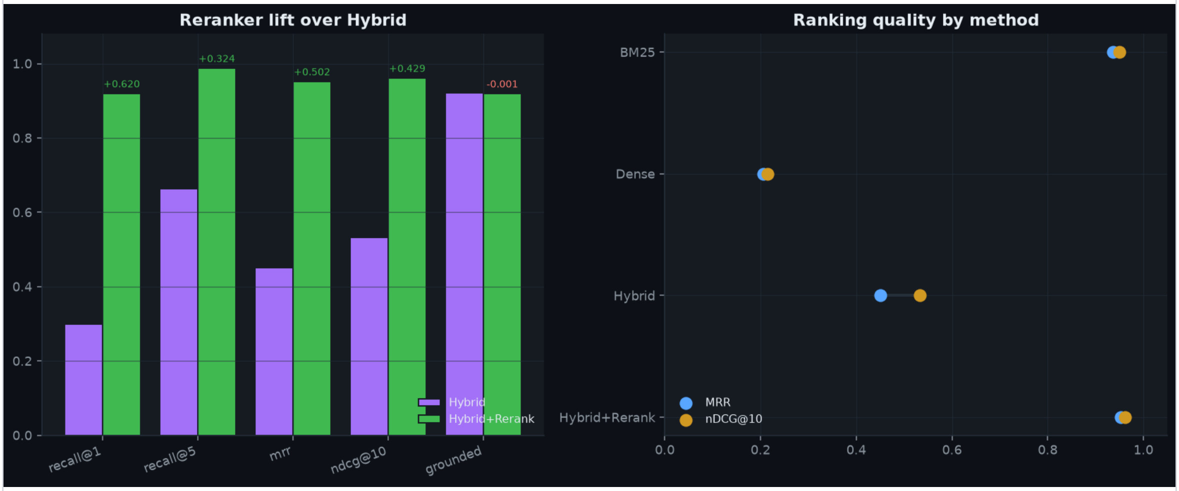

Figure 8.2. Reranker lift & ranking quality

8.3 Validation Checklist

Item	Status	Evidence
Automated tests	Available	28 tests committed in repository validation flow.
Benchmark results	Available	Benchmark markdown tables and result summaries were reviewed.
Screenshots	Available	4 screenshot asset(s) included in the repository evidence set.
Architecture note	Available	ARCHITECTURE.md documents design choices, scale assumptions, and trade-offs.

9. Operational Discussion

The final stack turned retrieval into a measurable engineering system. Hybrid retrieval with reranking improved top-rank quality while preserving grounded, citation-backed answers and deterministic offline reproducibility. The scorecards show Hybrid+Rerank leading both BM25 and dense-only retrieval on top-rank quality. Latency remains operationally plausible because the reranker only touches a bounded candidate set. Groundedness stays stable across retrieval modes because synthesis remains extractive and citation-aware. The analysis indicates that the repository is strongest where implementation, benchmark artefacts, and screenshots point to the same conclusion.

9.1 Constraints And Risks

Tenant leakage: Mitigated by applying tenant masks before scoring rather than filtering results after ranking. Dense recall weakness: Documented honestly and counter-balanced with hybrid fusion plus reranking. Scale growth: Addressed by a sharded ANN path and streaming corpus generation strategy.

These limitations do not invalidate the project. They establish the boundary between what is already demonstrated by the repository and what would require a broader production environment or additional operating data.

10. Closing Assessment

This project document concludes that Enterprise RAG Platform is best understood as a complete technical implementation supported by repository-native evidence. This brief documents a retrieval-augmented generation core built for enterprise knowledge use cases where accuracy, traceability, and tenant boundaries matter more than surface-level demo output.

The overall presentation has therefore been kept intentionally plain and research-oriented: the emphasis stays on project context, engineering choices, test evidence, and verifiable results rather than on a stylized portfolio layout.

10.1 Forward Work

- Swap the deterministic embedder for a production encoder behind the same interface.
- Add online evaluation slices for business unit, tenant tier, and corpus freshness.
- Move from in-memory retrieval to shard-aware ANN infrastructure for larger corpora.

11. Repository Appendix

11.1 Repository Evidence Inventory

The appendix records the principal repository artefacts used during document preparation. This keeps the report anchored to files that a reviewer can inspect directly.

Artefact	Category	Purpose
README.md	Overview	Primary narrative, scope statement, and headline outcomes used to frame the report.
ARCHITECTURE.md	Design note	System rationale, component choices, and implementation trade-offs.
benchmarks/latency.csv	Benchmark	Quantitative result or execution artefact used in the findings section.
benchmarks/results.json	Benchmark	Quantitative result or execution artefact used in the findings section.
benchmarks/retrieval_quality.csv	Benchmark	Quantitative result or execution artefact used in the findings section.
benchmarks/retrieval_quality.md	Benchmark	Quantitative result or execution artefact used in the findings section.
assets/quality_comparison.png	Screenshot	Visual validation surface referenced as 'Retrieval quality - 4 methods compared' in the document review.
assets/scorecard.png	Screenshot	Visual validation surface referenced as 'Evaluation scorecard' in the document review.
assets/latency_vs_k.png	Screenshot	Visual validation surface referenced as 'Query latency vs retrieval depth' in the document review.
assets/reranker_gain.png	Screenshot	Visual validation surface referenced as 'Reranker lift & ranking quality' in the document review.
tests/	Validation	Automated test suite containing 28 committed tests used to support implementation claims.

11.2 Internal References

1. README.md - primary repository overview and headline results.
2. ARCHITECTURE.md - system design, implementation rationale, and scale notes.
3. benchmarks/latency.csv - benchmark or result artefact used in the analysis section.

4. benchmarks/results.json - benchmark or result artefact used in the analysis section.
5. benchmarks/retrieval_quality.csv - benchmark or result artefact used in the analysis section.
6. benchmarks/retrieval_quality.md - benchmark or result artefact used in the analysis section.
7. tests/ - automated validation suite used to support implementation claims.
8. assets/latency_vs_k.png - generated screenshot evidence included in the report.